MATHEMATICS
Foundations of Algebra
Fall semester 2002

Text: Lecture notes (by Aaron Bertram) will be provided in class. They can also be obtained from: http://www.math.utah.edu/~bertram/4030/4030.pdf

When: M, W, F 2.00–2.50 P.M.

Where: JWB 333

Instructor: Prof. Christopher Hacon

Office: JWB 325

Email: hacon@math.utah.edu

Office hours: M 12:00-12:45, W 2:50-3:30, F 2:50-3:30.

Prerequisites: Multi-variable calculus and linear algebra are desirable. If you have not had a course on these subjects, please see me.

Grading: There will be six quizzes, two midterms, and a comprehensive final examination. Homework will be assigned but will not be collected. The quizzes and the midterms will be based on the homework. It is strongly recommended that you do all of the homework and seek help for any trouble with the homework or with understanding the material covered in class.

Homework will be assigned each Wednesday. There will be 6 quizzes. These will not be "announced", however they will be on a Wednesday, and they will cover material included in the homework assigned up to and including the previous Wednesday. Your lowest two quiz scores will be dropped and each of the remaining four quizzes will be worth 50 points. The midterms are scheduled on **Monday, Sept. 23** and on **Monday, Oct. 28**. Each midterm will be worth 100 points. The final exam (on **Monday, December 9, 1:00–3:00 P.M.**) will make up the remaining 200 points. Books, calculators and notes will not be allowed at the exams. Make up tests and exams will only be given for serious and documented reasons.
Totals: 200 + 200 + 200 = 600 possible points.

It is the Math Department policy, and mine as well, to grant any withdrawal request until the University deadline of October 18 (there is no tuition penalty if dropped by August 30).

ADA statement: The American with Disabilities Act requires that reasonable accommodations be provided for students with physical, sensory, cognitive, systemic, learning, and psychiatric disabilities. Please contact me at the beginning of the semester to discuss any such accommodations for the course.

Course outline: The course is divided into three parts:

1. A brief tour of number systems:
   a. Natural numbers (coumting) $\mathbb{N} = \{1, 2, 3, ...\}$, primes and induction.
   b. The ring of integers $\mathbb{Z} = \{..., -2, -1, 0, 1, 2, ...\}$.
   c. The field of rational numbers (fractions) $\mathbb{Q} = \{a/b\}$.
   d. The field of real numbers (decimal expansions) $\mathbb{R}$.
   f. The field of complex numbers $\mathbb{C}$.
   g. Number like systems: Quaternions and clock arithmetic.

2. Polynomials
   a. Long division, factoring and roots
   b. The fundamental theorem of algebra.
   c. Quadratic, cubic and quartic formulas.
   d. Using an algebraic number to build a new number system.
   e. Clock arithmetic for polynomials. Why we can't trisect an angle.

3. Matrices.
   a. "Groups" of invertible matrices (with examples).
   b. When vector spaces are algebras.
   c. Some information about the symmetric groups.
   d. Galois groups.
   e. Why there is no good formula for finding roots of all polynomials.